

Physics-Guided Latent Diffusion for Lensless Depth from PSF Stacks

Donggeon Bae Hojin Jang — 42-plane PSF stack, 66k train / 6k test, depth-guided all-in-focus reconstruction

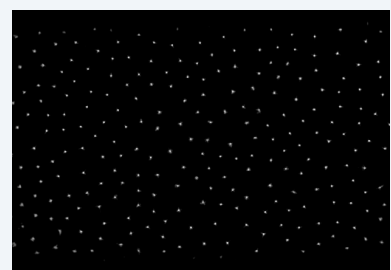
- A lensless measurement mixes RGB radiance through depth-dependent PSFs.
- We estimate depth from one raw measurement and a calibrated 42-plane PSF stack.
- The central cue is focus variation across PSF-Wiener deconvolution planes.

Data and Forward Physics

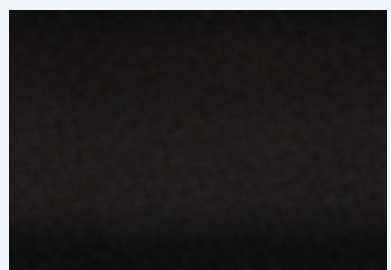
Training samples contain RGB ground truth, normalized depth labels, and a calibrated 42-plane PSF stack. Lensless raw inputs are synthesized by depth-sliced convolution:

$$y = \sum_{k=1}^{42} (w_k(d) \odot x) + b_k + \eta,$$

where $w_k(d)$ softly bins each pixel depth into neighboring PSF planes.



PSF plane



lensless raw



GT RGB



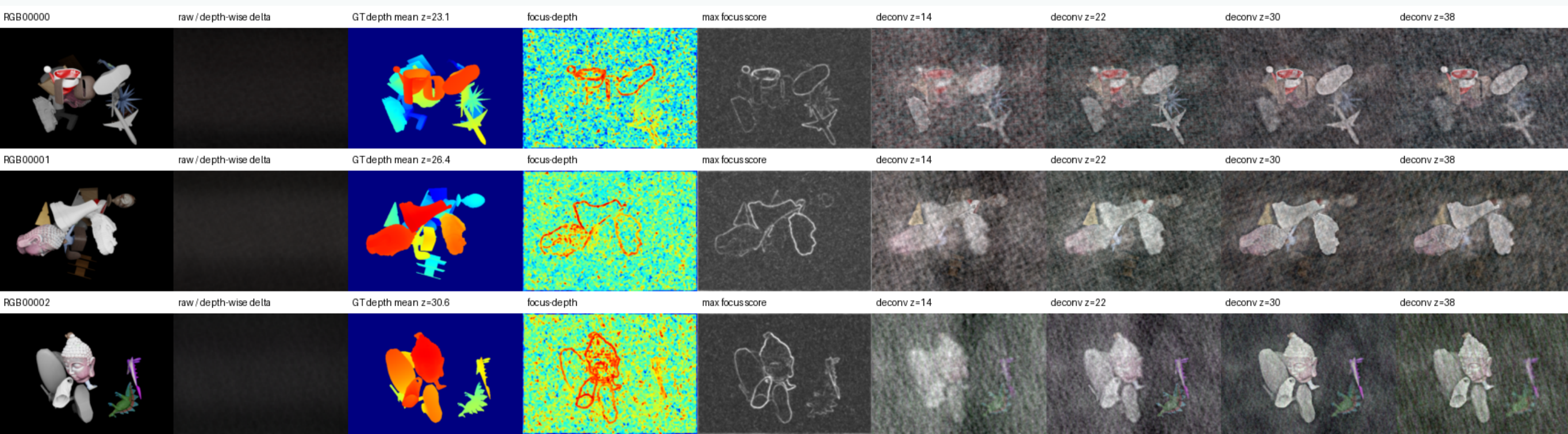
GT depth

PSF-Stack Deconvolution

For each depth plane, we compute a Wiener inverse:

$$\hat{x}_k = \mathcal{F}^{-1} \left(\frac{\overline{H_k} Y}{|H_k|^2 + \lambda_k} \right).$$

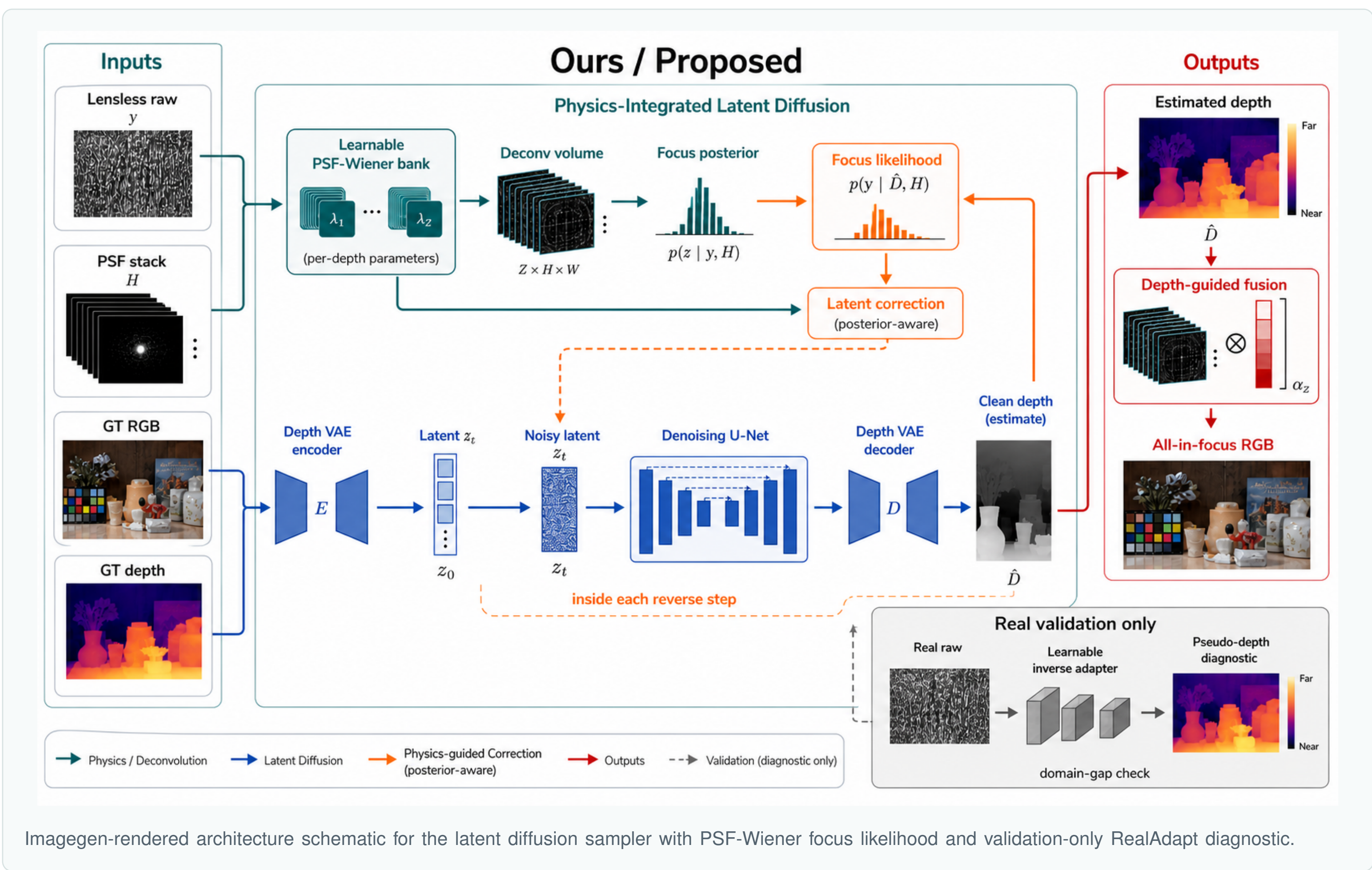
Early depth planes produce weak or unstable inverses, so the poster visualizes only later useful planes. The four panels below use $z = \{14, 22, 30, 38\}$.



Mid/late deconvolution planes reveal object contours and depth-dependent focus without unstable early-plane inverses.

Ours is the proposed clean-posterior latent diffusion model. Each reverse step:

1. predicts a clean depth latent z_0 from noisy z_t ;
2. decodes z_0 to metric depth;
3. evaluates focus likelihood from the PSF-Wiener deconvolution stack;
4. backpropagates the focus residual into z_t ;
5. repeats two short posterior refinements, then continues the DDIM transition.



Imagegen-rendered architecture schematic for the latent diffusion sampler with PSF-Wiener focus likelihood and validation-only RealAdapt diagnostic.

Why Diffusion?

Diffusion treats depth as a latent variable sampled under two forces:

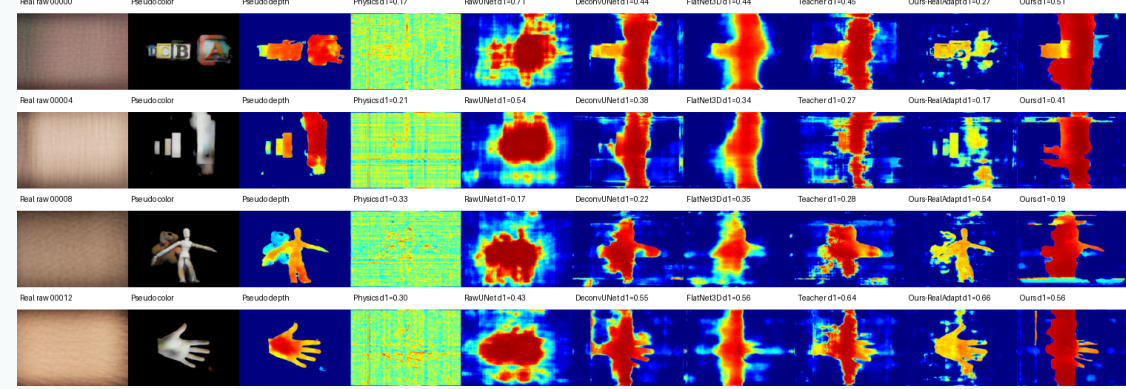
$$\nabla_{z_t} \log p(z_t | y) \approx \nabla_{z_t} \log p_D(z_t) - \alpha_t \nabla_{z_t} \mathcal{L}_{\text{focus}}(D(z_t), y).$$

The learned denoiser supplies a depth prior; the PSF-Wiener focus term pulls samples toward depths that explain the lensless measurement. RGB restoration is then produced by depth-weighted deconvolution-plane fusion.

We evaluate foreground depth metrics on the 6,000-sample test set. The final **Ours** row is selected by method structure: physics is applied inside the latent reverse process.

Method	Eval	δ_1	δ_2	MAE
Physics focus	6k	0.391	0.816	0.214
Raw U-Net, 66k	6k	0.436	0.753	0.204
Deconv U-Net, 66k	6k	0.865	0.949	0.050
Deconv U-Net + affine	6k	0.881	0.968	0.046
Earlier integrated LDM	6k	0.878	0.948	0.059
Ours / Proposed	6k	0.896	0.932	0.050

Real pseudo diagnostic. Learnable deconvolution adaptation trained for 30 epochs on measured captures gives 20250505 MAE 0.264 vs. fixed DeconvUNet 0.308, and 20250527-subset MAE 0.177 vs. physics focus 0.291.

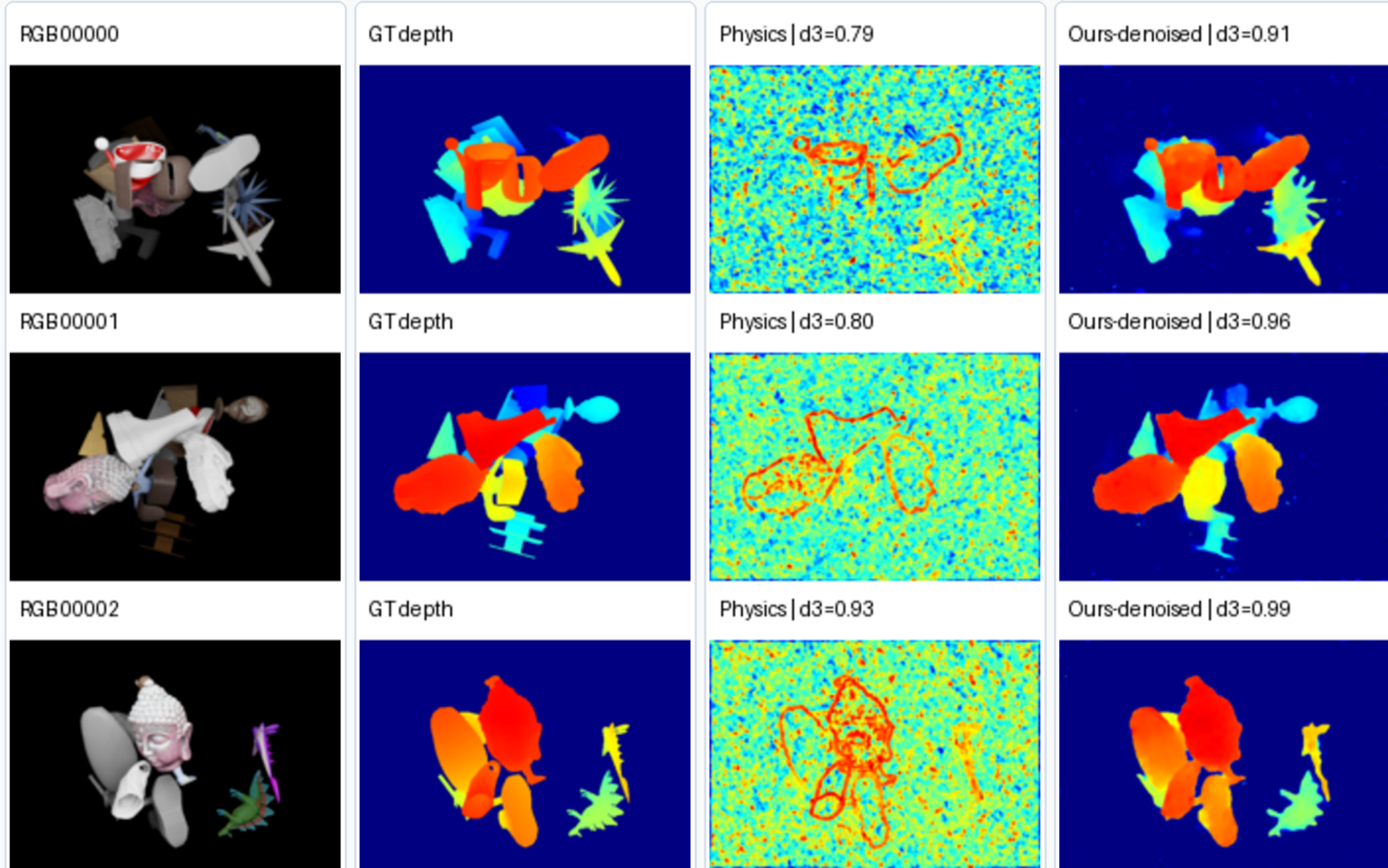


Real validation uses pseudo-depth labels; it is a domain-gap diagnostic, not measured GT.

Qualitative Results

Final depth estimation results

Denoised qualitative comparison: RGB input, GT depth, physics focus baseline, and Ours-denoised depth.



Denoised Ours panel from the clean-posterior refinement probe; full-6k metrics are reported separately in the table.

Final visual panel: RGB, GT depth, physics focus baseline, and denoised Ours depth only.

Findings

- Raw input alone is weak; PSF-Wiener deconvolution is the useful depth representation.
- Ours improves strict depth and visual denoising, but still trails direct supervised regression on loose δ_1 .
- Real captures need learnable deconvolution: a learned Wiener bank plus artifact adapter reduces pseudo-label error and suppresses fixed-inverse artifacts.
- The remaining gap is mainly calibration and boundary ambiguity, not total depth failure.

Takeaway

PSF-stack deconvolution is the right representation for lensless depth. **Ours** demonstrates how to move the same optical physics inside a latent diffusion posterior sampler. Future work should tune sigma-aware clean-latent correction and plane-to-depth calibration.

Final Ours metric full-6k: $\delta_1 = 0.896$, $\delta_2 = 0.932$, MAE = 0.050.

Real-adapt metric 20250505 pseudo MAE = 0.264; 20250527 subset pseudo MAE = 0.177.